VHDL Instantiation:

```
ram1024x4nr_inst:  SB_RAM1024x4NR
generic map (
INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
      )
port map (
        RDATA => RDATA_c,
        RADDR => RADDR_c,
        RCLKN => RCLKN_c,
        RCLKE => RCLKE_c,
        RE => RE_c,
        WADDR => WADDR_c,
        WCLK=> WCLK_c,
        WCLKE => WCLKE_c,
        WDATA => WDATA_c,
        WE => WE_c
      );
```

SB_RAM1024x4NW

SB_RAM1024x4NW // Negative edged Write Clock – i.e. WCLKN
(RDATA, RCLK, RCLKE, RE, RADDR, WCLKN, WCLKE, WE, WADDR, MASK, WDATA);

Verilog Instantiation:

```
SB_RAM1024x4NW   ram1024x4nw_inst (
       .RDATA(RDATA_c[3:0]),
       .RADDR(RADDR_c[9:0]),
       .RCLK(RCLK_c),
       .RCLKE(RCLKE_c),
       .RE(RE_c),
       .WADDR(WADDR_c[3:0]),
       .WCLKN(WCLKN_c),
       .WCLKE(WCLKE_c),
       .WDATA(WDATA_c[9:0]),
       .WE(WE_c)
);
defparam ram1024x4_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

VHDL Instantiation:

```
ram1024x4nw_inst :  SB_RAM1024x4NW
generic map (
INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
```

```
INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
      )
port map (
        RDATA => RDATA_c,
        RADDR => RADDR_c,
        RCLK => RCLK_c,
        RCLKE => RCLKE_c,
        RE => RE_c,
        WADDR => WADDR_c,
        WCLKN=> WCLKN_c,
        WCLKE => WCLKE_c,
        WDATA => WDATA_c,
        WE => WE_c
      );
```

SB_RAM1024x4NRNW

SB_RAM1024x4NRNW **// Negative edged Read and Write – i.e. RCLKN WRCKLN**
(RDATA, RCLKN, RCLKE, RE, RADDR, WCLKN, WCLKE, WE, WADDR, MASK, WDATA);

Verilog Instantiation:

```
SB_RAM1024x4NRNW   ram1024x4nrnw_inst (
      .RDATA(RDATA_c[3:0]),
      .RADDR(RADDR_c[9:0]),
      .RCLKN(RCLK_c),
      .RCLKE(RCLKE_c),
      .RE(RE_c),
      .WADDR(WADDR_c[3:0]),
      .WCLKN(WCLK_c),
      .WCLKE(WCLKE_c),
      .WDATA(WDATA_c[9:0]),
      .WE(WE_c)
);
defparam ram1024x4nrnw_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

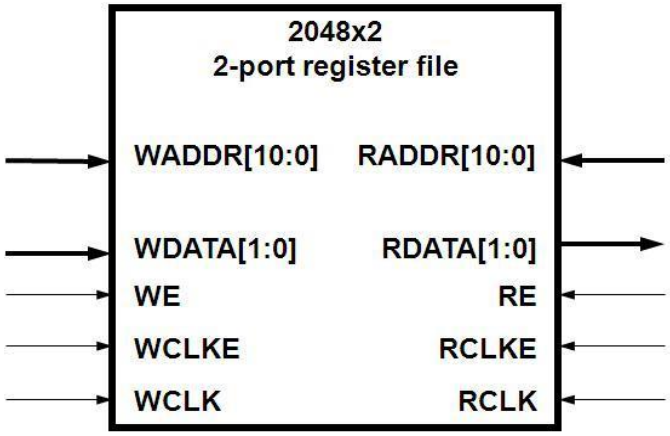
```
defparam ram1024x4nrnw_inst.INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nrnw_inst.INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

VHDL Instantiation:

```
ram1024x4nrnw_inst :  SB_RAM1024x4NRNW
generic map (
INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
      )
```

```
port map (
        RDATA => RDATA_c,
        RADDR => RADDR_c,
        RCLKN => RCLKN_c,
        RCLKE => RCLKE_c,
        RE => RE_c,
        WADDR => WADDR_c,
        WCLKN=> WCLKN_c,
        WCLKE => WCLKE_c,
        WDATA => WDATA_c,
        WE => WE_c
      );
```

SB_RAM2048x2



SB_RAM2048x2

The following modules are the complete list of SB_RAM2048x2 based primitives

SB_RAM2048x2

SB_RAM2048x2 **//Posedge clock RCLK WCLK**
(RDATA, RCLK, RCLKE, RE, RADDR, WCLK, WCLKE, WE, WADDR, MASK, WDATA);

Verilog Instantiation:

```
SB_RAM2048x2   ram2048x2_inst (
      .RDATA(RDATA_c[1:0]),
      .RADDR(RADDR_c[10:0]),
      .RCLK(RCLK_c),
      .RCLKE(RCLKE_c),
      .RE(RE_c),
      .WADDR(WADDR_c[10:0]),
      .WCLK(WCLK_c),
      .WCLKE(WCLKE_c),
      .WDATA(WDATA_c[1:0]),
      .WE(WE_c)
);
defparam ram2048x2_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

```
defparam ram2048x2_inst .INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2_inst .INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

VHDL Instantiation:

```
Ram2048x2_inst :  SB_RAM2048x2
generic map (
INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
      )
port map (
        RDATA => RDATA_c,
        RADDR => RADDR_c,
        RCLK => RCLK_c,
        RCLKE => RCLKE_c,
        RE => RE_c,
        WADDR => WADDR_c,
        WCLK=> WCLK_c,
        WCLKE => WCLKE_c,
```

```
        WDATA => WDATA_c,
        WE => WE_c
      );
```

SB_RAM2048x2NR

SB_RAM2048x2NR // Negative edged Read Clock – i.e. RCLKN
(RDATA, RCLKN, RCLKE, RE, RADDR, WCLK, WCLKE, WE, WADDR, MASK, WDATA);

```
SB_RAM2048x2NR   ram2048x2nr_inst (
      .RDATA(RDATA_c[1:0]),
      .RADDR(RADDR_c[10:0]),
      .RCLKN(RCLKN_c),
      .RCLKE(RCLKE_c),
      .RE(RE_c),
      .WADDR(WADDR_c[10:0]),
      .WCLK(WCLK_c),
      .WCLKE(WCLKE_c),
      .WDATA(WDATA_c[1:0]),
      .WE(WE_c)
);
defparam ram2048x2nr_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nr_inst .INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

VHDL Instantiation:

```
ram2048x2nr_inst :  SB_RAM2048x2NR
generic map (
INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
```

```
INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
      )
port map (
        RDATA => RDATA_c,
        RADDR => RADDR_c,
        RCLKN => RCLKN_c,
        RCLKE => RCLKE_c,
        RE => RE_c,
        WADDR => WADDR_c,
        WCLK=> WCLK_c,
        WCLKE => WCLKE_c,
        WDATA => WDATA_c,
        WE => WE_c
      );
```

SB_RAM2048x2NW

SB_RAM2048x2NW **// Negative edged Write Clock – i.e. WCLKN**
(RDATA, RCLK, RCLKE, RE, RADDR, WCLKN, WCLKE, WE, WADDR, MASK, WDATA);

```
SB_RAM2048x2NW   ram2048x2nw_inst (
      .RDATA(RDATA_c[1:0]),
      .RADDR(RADDR_c[10:0]),
      .RCLK(RCLK_c),
      .RCLKE(RCLKE_c),
      .RE(RE_c),
      .WADDR(WADDR_c[10:0]),
      .WCLKN(WCLKN_c),
      .WCLKE(WCLKE_c),
      .WDATA(WDATA_c[1:0]),
      .WE(WE_c)
);
defparam ram2048x2nw_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

```
defparam ram2048x2nw_inst .INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram2048x2nw_inst .INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

VHDL Instantiation:

```
ram2048x2nw_inst:  SB_RAM2048x2NW
generic map (
INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
```